

Q12 Labour members (proportion p) $\rightarrow$ $P(L) = p$
 $P(L^c) = 1 - p$

~~P~~ ~~R~~ R : Change their minds

L : Labour members

L^c : Conservative

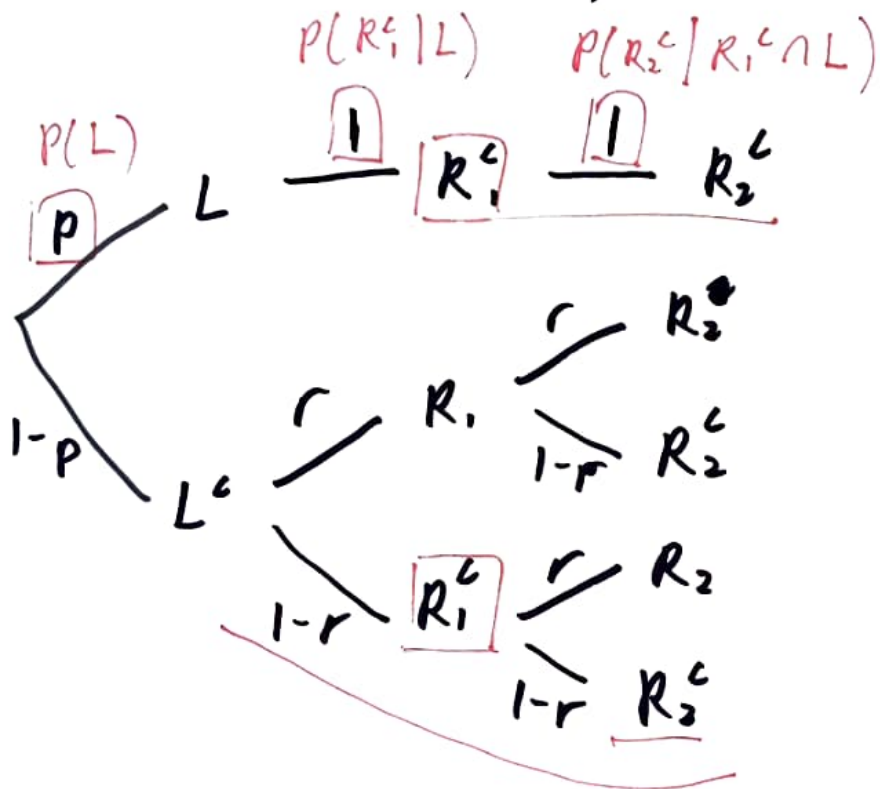
$$P(R_i | L) = 0 \Rightarrow P(R_i^c | L) = 1$$
$$P(R_i | L^c) = r \Rightarrow P(R_i^c | L^c) = 1 - r$$

~~Describe~~ Scene: Vote 1 $\xrightarrow{R_1^c}$ Vote 2 $\xrightarrow{R_2^c}$ Vote 3

Event: $R_1, R_1^c, R_2, R_2^c, L, L^c$

Find $P(R_2^c | R_1^c)$

$$P(R_2^c | R_1^c) = \frac{P(R_2^c \cap R_1^c)}{P(R_1^c)}$$



Thm 6.2 (3)

$$\cancel{P(A|B \cap C)} = \cancel{\frac{P(A \cap B | C)}{P(B | C)}}$$

$$P(A \cap B | C) = P(A | B \cap C) P(B | C) P(C)$$

$$P(R_2^c | R_1^c) = \frac{P(R_2^c \cap R_1^c \cap L) + P(R_2^c \cap R_1^c \cap L^c)}{P(L \cap R_1^c) + P(L^c \cap R_1^c)}$$

$$= \frac{p \cdot 1 \cdot 1 + (1-p)(1-r)(1-r)}{p \cdot 1 + (1-p)(1-r)}$$

= ...

$$\frac{P(L)}{p} \cdot \frac{P(R_1^c | L)}{1} \cdot \frac{P(R_2^c | R_1^c \cap L)}{P(R_2^c | L)}$$

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1

$$R_1^c \text{ and } R_2^c \text{ are indep.} \Rightarrow P(R_2^c | R_1^c \cap L) = P(R_2^c | L)$$